

Machine Learning

Week 3 Lecture

Dr. Muhammad Ahmed Raza

Contents

- Intro to Support Vector Machines
- The objective function
- Kernel trick

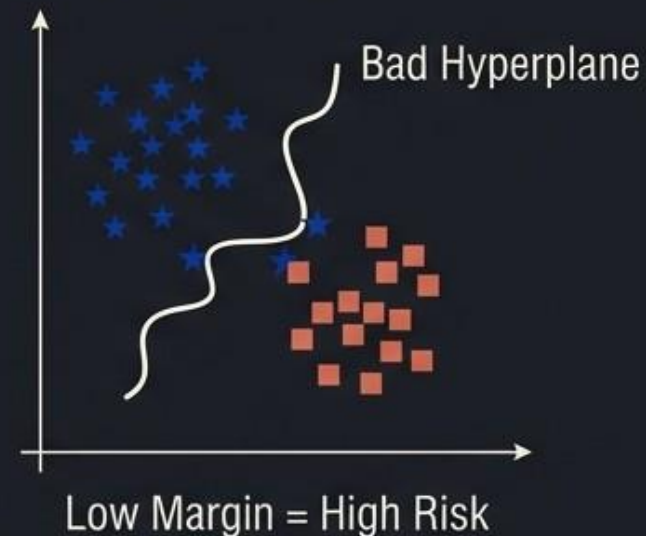
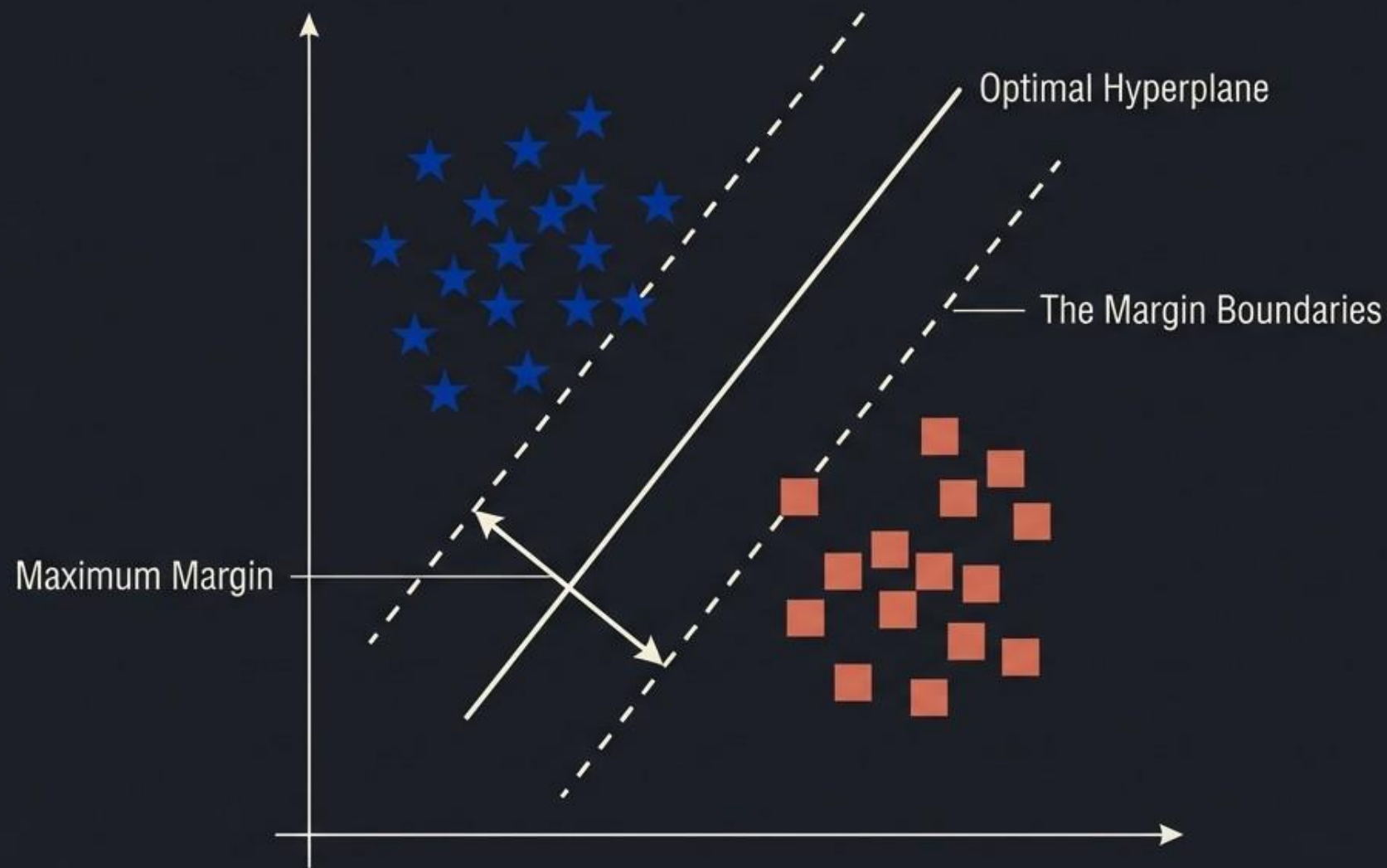
Intro to SVMs

- Support vector machines are non-probabilistic binary linear classifiers.
- The use of basis functions and the kernel trick mitigates the constraint of the SVM being a linear classifier – in fact SVMs are particularly associated with the kernel trick.
- Only a subset of data-points are required to define the SVM classifier - these points are called support vectors.
- SVMs are very popular classifiers and applications include –
 - text classification
 - outlier detection
 - face detection, etc

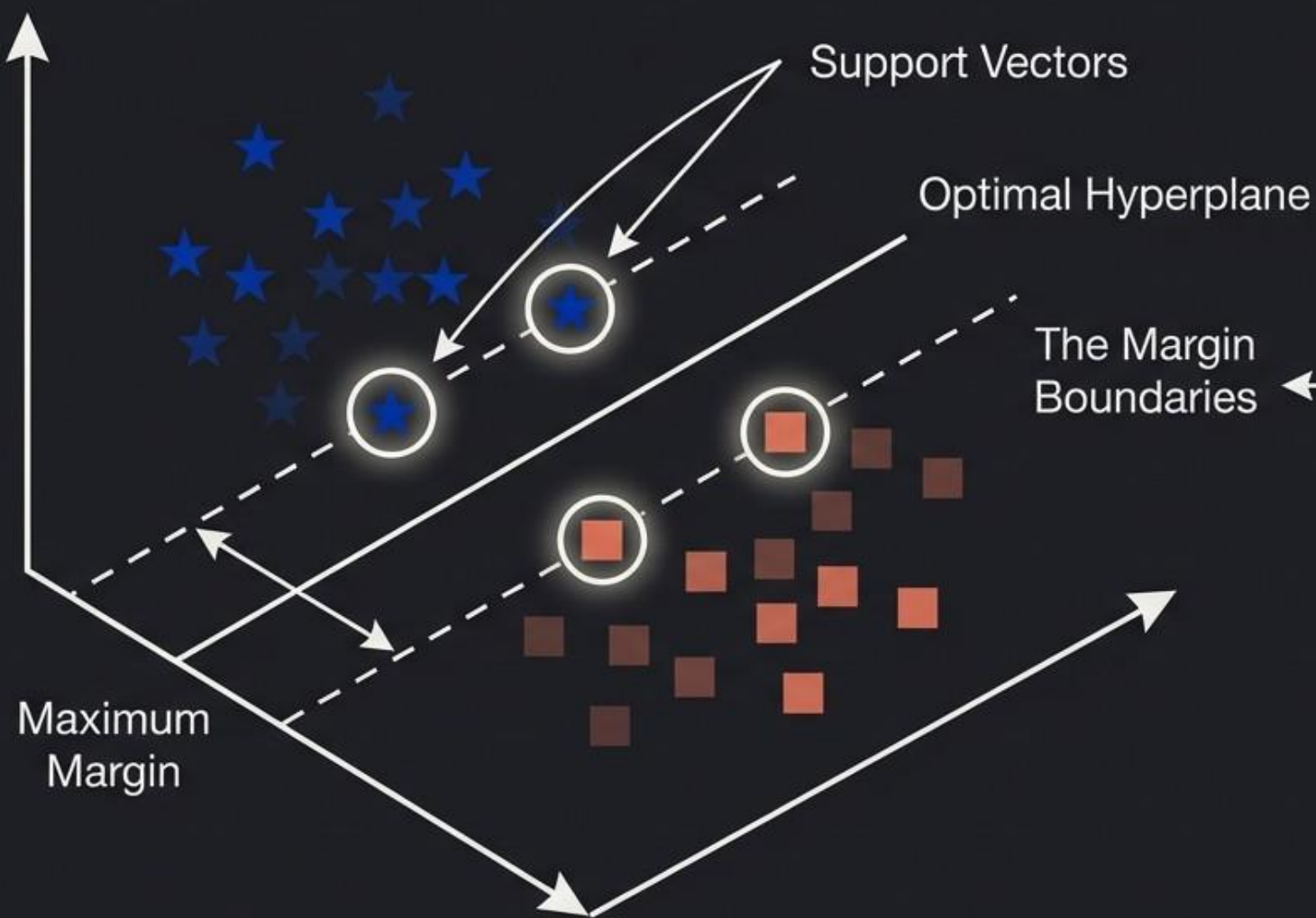
Intro to SVMs

- SVM is primarily a two-class* classifier that maximises the width of the margin between classes, that is, the empty area around the decision boundary defined by the distance to the nearest training patterns.
- *It is feasible to extend the system to more than 2 classes
- Notations:
 - In $y(x) = w^T x + b$
 - w defines orientation
 - b shifts the plane
 - The decision boundary is a hyperplane

The Intuition of the Widest Street



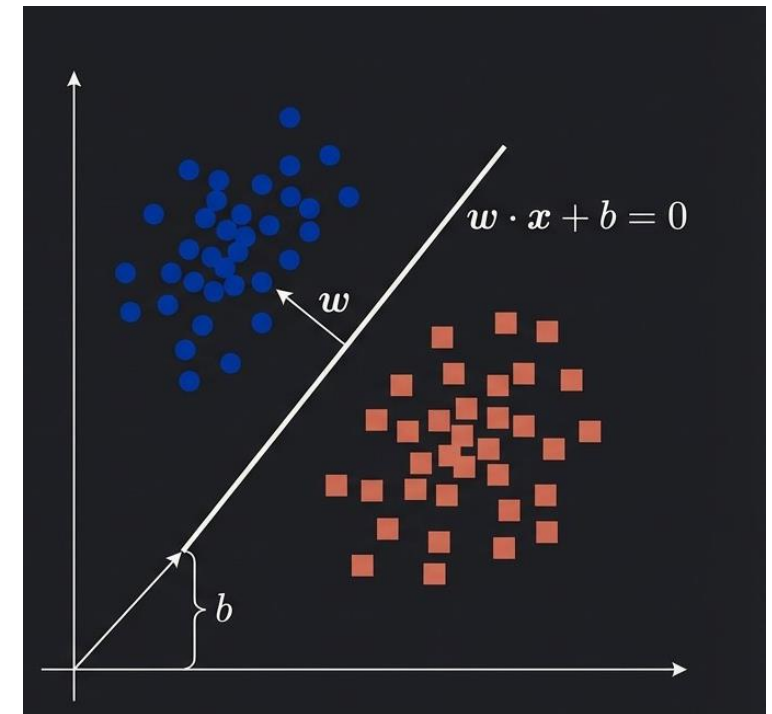
The Critical Few: Support Vectors



Radon's Theorem implies the solution depends only on these intersection points. If you remove the faded points, the optimal hyperplane does not move.

The Separable Case

- There are two classes which are assumed (for now) to be linearly separable.
- Training data $x_1, \dots, x_n$ with corresponding targets with $y_i \in \{-1, 1\}$.
- We consider a classification rule of the form of the form
$$h(x) = \text{sign}(w^T x + b)$$
$$h(x) = \text{sign}(y(x))$$
- where $y(x) = w^T x + b$.



The Separable Case

- Note we can re-scale (w, b) without changing the decision boundary (for $c > 0$).

For decision boundary

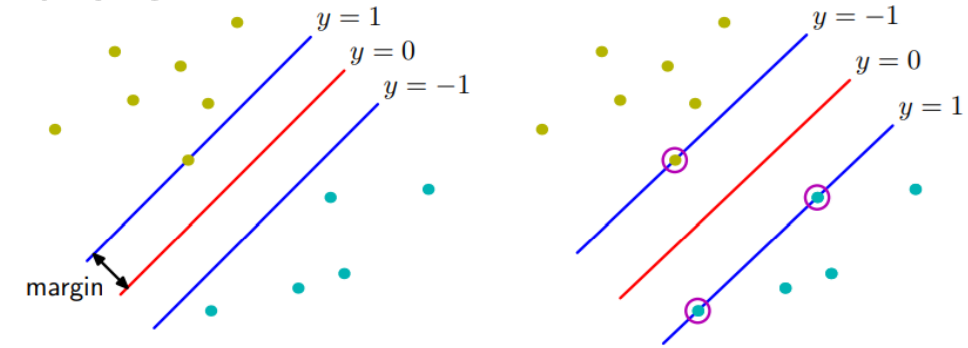
$$(cw)^T x + cb = c(w^T x + b)$$

$$c(w^T x + b) = 0$$

Classifier

$$\text{sign}(w^T x + b)$$

$$(w, b) \quad \text{and} \quad (cw, cb)$$



- Therefore choose (w, b) so that training points closest to boundary satisfy $y(x) = \pm 1$
- Let x_1 be closest point from class with $y_1 = -1$ so that $w^T x_1 + b = -1$. And let x_2 be closest point from class with $y_2 = 1$ so that $w^T x_2 + b = 1$.

What is margin?

- Distance from a point to a hyperplane.
 - Mathematically:
 - the perpendicular distance of a point x from the hyperplane, $w^T x + b = 0$, is:

$$\frac{|w^T x + b|}{\|w\|}$$

- For the closest point:

$$|w^T x + b| = 1$$

$$\frac{1}{\|w\|}$$

- For both sides (distance between two margins):

$$\frac{2}{\|w\|}$$

The objective function

- $\max \frac{1}{\|w\|^2} \Leftrightarrow \min \|w\|^2 \text{ s.t. } y_i(w^T x_i + b) \geq 1$
- Intuition:
 - Objective function: maximizes margin
 - Constraint: classification + scale
 - Classification: $y_i(w^T x + b) > 0$
 - Scale: closest point set to 1
- Summary:
 - Because we can rescale (w, b) freely, SVM fixes the scale so that the closest points equal ± 1 — which makes maximizing the margin exactly the same as minimizing $\|w\|^2$.